

A NOTE ON THE BOUND $g(n) \ll_\varepsilon n^{10/33+\varepsilon}$

CHATGPT

ABSTRACT. For a positive integer n , let $g(n)$ denote the least scale on which a multiplicative Sidon subset of $[n]$ can be made to meet every interval. Using the currently available exceptional-set estimate for the prime number theorem in short intervals due to Gafni and Tao, we prove

$$g(n) \ll n^{10/33+o(1)}.$$

The proof uses primes to cover the prime-rich blocks and then applies a local lemma argument to fill the remaining prime-poor blocks. Under the Riemann hypothesis, the same method gives

$$g(n) \ll n^{1/4+o(1)}.$$

1. INTRODUCTION AND EXTERNAL INPUT

Throughout the paper, we write $\mathbb{N} := \{1, 2, 3, \dots\}$. A set $A \subseteq \mathbb{N}$ is called a *multiplicative Sidon set* if all products aa' with $a, a' \in A$ and $a \leq a'$ are distinct.

Multiplicative Sidon sets have been studied since the work of Erdős [4, 5]. In particular, the maximum size of a multiplicative Sidon subset of $[n] := \{1, \dots, n\}$ is

$$\pi(n) + \Theta\left(\frac{n^{3/4}}{(\log n)^{3/2}}\right).$$

Thus the extremal theory of multiplicative Sidon sets is quite well understood from the point of view of cardinality. The corresponding spacing problem, however, is much less developed.

In his 2001 list of unsolved problems, Sárközy [7, Problem 34] asked the following problem.

Problem 1.1 (Sárközy [7]). *How small can one make the maximal gap between the consecutive elements of a multiplicative Sidon set selected from $\{1, 2, \dots, n\}$? Does there exist for all n a multiplicative Sidon set $A \subseteq \{1, 2, \dots, n\}$ so that $A \cap [m, m + \sqrt{n}] \neq \emptyset$ for all $1 \leq m < m + \sqrt{n} \leq n$?*

Sárközy observed that a positive answer to the second question would follow, for example, from a prime-gap estimate of the shape $p_{i+1} - p_i = o(p_i^{1/2})$. Such a bound is far beyond what is currently known.

To encode Problem 1.1 quantitatively, we use the following function.

Definition 1.2. For $n \in \mathbb{N}$, define $g(n)$ to be the infimum of all real numbers $L \geq 0$ such that there exists a multiplicative Sidon set $A \subseteq [n] := \{1, 2, \dots, n\}$ with the property that

$$A \cap [x, x + L] \neq \emptyset \quad \text{for every } x \in \mathbb{R} \text{ with } [x, x + L] \subseteq [1, n].$$

A natural upper-bound benchmark is provided by the primes: since the set of primes is multiplicatively Sidon, any result on primes in short intervals gives an upper bound for $g(n)$. For example, the theorem of Baker, Harman and Pintz [2] implies

$$g(n) \ll n^{21/40}.$$

In the opposite direction, the cardinality bound above, together with the prime number theorem and the pigeonhole principle, gives

$$g(n) \geq (1 + o(1)) \log n.$$

Recently, van Doorn, Monticone and Tang [3] gave an affirmative answer to Sárközy's explicit $\sqrt{n}$ -question and proved the power-saving bound

$$g(n) \ll_{\varepsilon} n^{\rho+\varepsilon}, \quad \rho := \frac{13 - \sqrt{69}}{10} < 0.47.$$

The purpose of the present note is to show that the exceptional-set estimates for primes in short intervals due to Gafni and Tao lead to the stronger bound

$$g(n) \ll_{\varepsilon} n^{10/33+\varepsilon}.$$

We shall use the following consequence of the work of Gafni and Tao on the exceptional set in the prime number theorem in short intervals.

Theorem 1.3 (Gafni–Tao, consequence of the exceptional-set bound). *Let $10/33 < \theta < 1$. For every $\eta > 0$ and all sufficiently large X ,*

$$\left| \left\{ x \in [X, 2X] : \left| \sum_{x < m \leq x+x^{\theta}} \Lambda(m) - x^{\theta} \right| \geq \frac{1}{2} x^{\theta} \right\} \right| \ll_{\theta, \eta} X^{10/11+\eta},$$

where $|\cdot|$ denotes Lebesgue measure.

Remark 1.4 (Verification of Theorem 1.3). We spell out the finite calculation needed from Gafni–Tao. Let

$$\theta_0 := \frac{10}{33}, \quad c_0 := \frac{1}{1 - \theta_0} = \frac{33}{23}.$$

Gafni–Tao prove that, for every fixed $0 < \theta < 1$,

$$\mu(\theta) \leq \inf_{\xi > 0} \sup_{\substack{0 \leq \sigma < 1 \\ A(\sigma) \geq (1-\theta)^{-1}-\xi}} ((1-\theta)(1-\sigma)A(\sigma) + 2\sigma - 1).$$

Let $\tilde{A}(\sigma)$ be the piecewise upper bound for $A(\sigma)$ given in Table 1 of Gafni–Tao, together with

$$\tilde{A}(\sigma) = \frac{1}{1-\sigma} \quad (0 \leq \sigma \leq 1/2).$$

Set

$$F(\sigma) := \frac{23}{33}(1-\sigma)\tilde{A}(\sigma) + 2\sigma - 1.$$

It suffices to verify that

$$F(\sigma) \leq \frac{10}{11} \quad \text{whenever } \tilde{A}(\sigma) \geq \frac{33}{23}.$$

The condition $\tilde{A}(\sigma) \geq 33/23$ restricts the last relevant Table 1 entry

$$\tilde{A}(\sigma) = \frac{3}{10\sigma - 7} \quad \left(\frac{9}{10} < \sigma \leq \frac{31}{34} \right)$$

to $\sigma \leq 10/11$. No later Table 1 entry contributes. Taking closures of the relevant ranges, which can only enlarge the suprema, direct differentiation of the corresponding rational functions gives

the following finite check:

range of σ	$\tilde{A}(\sigma)$	$\sup F(\sigma)$
$\left[\frac{10}{33}, \frac{1}{2} \right]$	$\frac{1}{1-\sigma}$	$\frac{23}{33}$
$\left[\frac{1}{2}, \frac{7}{10} \right]$	$\frac{3}{2-\sigma}$	$\frac{631}{715}$
$\left[\frac{7}{10}, \frac{19}{25} \right]$	$\frac{15}{3+5\sigma}$	$\frac{4156}{4675}$
$\left[\frac{19}{25}, \frac{127}{167} \right]$	$\frac{9}{8\sigma-2}$	$\frac{4156}{4675}$
$\left[\frac{127}{167}, \frac{13}{17} \right]$	$\frac{15}{13\sigma-3}$	$\frac{1625}{1837}$
$\left[\frac{13}{17}, \frac{17}{22} \right]$	$\frac{6}{5\sigma-1}$	$\frac{985}{1122}$
$\left[\frac{17}{22}, \frac{41}{53} \right]$	$\frac{2}{9\sigma-6}$	$\frac{608}{693}$
$\left[\frac{41}{53}, \frac{7}{9} \right]$	$\frac{9}{7\sigma-1}$	$\frac{1721}{1980}$
$\left[\frac{7}{9}, \frac{1867}{2347} \right]$	$\frac{9}{8(2\sigma-1)}$	$\frac{1721}{1980}$
$\left[\frac{1867}{2347}, \frac{7}{8} \right]$	$\frac{3}{2\sigma}$	$\frac{277}{308}$
$\left[\frac{7}{8}, \frac{279}{314} \right]$	$\frac{3}{10\sigma-7}$	$\frac{920849}{1022384}$
$\left[\frac{279}{314}, \frac{9}{10} \right]$	$\frac{24}{30\sigma-11}$	$\frac{199}{220}$
$\left[\frac{9}{10}, \frac{10}{11} \right]$	$\frac{3}{10\sigma-7}$	$\frac{10}{11}$

Hence

$$F(\sigma) \leq \frac{10}{11} \quad \text{on the range } \tilde{A}(\sigma) \geq \frac{33}{23}.$$

Now fix $\theta > 10/33$, and put $c_\theta = (1 - \theta)^{-1}$. Since $c_\theta > c_0$, choose $0 < \xi < c_\theta - c_0$. If

$$A(\sigma) \geq c_\theta - \xi,$$

then

$$\tilde{A}(\sigma) \geq A(\sigma) > c_0.$$

Using $A(\sigma) \leq \tilde{A}(\sigma)$ and $1 - \theta < 1 - \theta_0 = 23/33$, we obtain

$$\begin{aligned} (1 - \theta)(1 - \sigma)A(\sigma) + 2\sigma - 1 &\leq \frac{23}{33}(1 - \sigma)\tilde{A}(\sigma) + 2\sigma - 1 \\ &\leq \frac{10}{11}. \end{aligned}$$

Therefore the Gafni–Tao bound gives

$$\mu(\theta) \leq \frac{10}{11} \quad (\theta > 10/33).$$

Taking $\delta = 1/2$ in the definition of the exceptional-set exponent gives Theorem 1.3.

We shall also use the following conditional consequence of the same exceptional-set framework.

Theorem 1.5. *Assume the Riemann hypothesis. Let $0 < \theta \leq 1/2$. For every $\eta > 0$ and all sufficiently large X ,*

$$\left| \left\{ x \in [X, 2X] : \left| \sum_{x < m \leq x+x^\theta} \Lambda(m) - x^\theta \right| \geq \frac{1}{2} x^\theta \right\} \right| \ll_{\theta, \eta} X^{1-\theta+\eta},$$

where $|\cdot|$ denotes Lebesgue measure.

We also use a standard elementary estimate for prime powers in very short intervals.

Lemma 1.6 (Prime powers are negligible below the square-root scale). *Fix $0 < \theta < 1/2$. Uniformly for $x \geq 2$,*

$$\sum_{\substack{x < m \leq x+x^\theta \\ m=p^k, k \geq 2}} \Lambda(m) \ll_\theta (\log x)^2.$$

Proof. For a fixed $k \geq 2$, the number of prime powers $p^k \in (x, x+x^\theta]$ is at most

$$(x+x^\theta)^{1/k} - x^{1/k} + 1 \ll x^{\theta+1/k-1} + 1.$$

Summing over $2 \leq k \leq \log_2(2x)$ gives

$$\ll x^{\theta-1/2} \log x + \log x \ll_\theta \log x$$

prime powers. Each contributes at most $O(\log x)$ to Λ , and the claim follows. $\square$

2. TWO COMBINATORIAL LEMMAS

The first lemma is a convenient local-lemma form tailored to our selection problem.

Lemma 2.1 (A variable local lemma). *Let $X_1, \dots, X_N$ be mutually independent random variables, and let $\mathcal{E}$ be a finite family of bad events. Suppose that every $E \in \mathcal{E}$ is determined by at least one and at most r of the variables, where $r \geq 1$. For a variable X_i , write $i \in \text{supp}(E)$ if E depends on X_i . If*

$$\sum_{E \in \mathcal{E}: i \in \text{supp}(E)} \mathbb{P}(E) \leq \frac{1}{8r} \quad (1 \leq i \leq N),$$

then with positive probability none of the events in $\mathcal{E}$ occurs.

Proof. We use the asymmetric Lovász local lemma. Put

$$x_E := 2\mathbb{P}(E) \quad (E \in \mathcal{E}).$$

Since every event has non-empty support, for any $E \in \mathcal{E}$ and any $i \in \text{supp}(E)$ we have

$$\mathbb{P}(E) \leq \sum_{F: i \in \text{supp}(F)} \mathbb{P}(F) \leq \frac{1}{8r}.$$

Thus $0 \leq x_E \leq 1/(4r) < 1$.

For the dependency graph, join two distinct events when they share at least one underlying variable. For any fixed bad event E ,

$$\sum_{F \sim E} x_F \leq 2 \sum_{i \in \text{supp}(E)} \sum_{F: i \in \text{supp}(F)} \mathbb{P}(F) \leq 2r \cdot \frac{1}{8r} = \frac{1}{4}.$$

Therefore

$$\prod_{F \sim E} (1 - x_F) \geq 1 - \sum_{F \sim E} x_F \geq \frac{3}{4},$$

and hence

$$x_E \prod_{F \sim E} (1 - x_F) \geq \frac{3}{2} \mathbb{P}(E) \geq \mathbb{P}(E).$$

The asymmetric Lovász local lemma now gives a positive probability of avoiding all bad events. $\square$

The second lemma constructs a multiplicative Sidon transversal from a sparse family of blocks.

Lemma 2.2 (Sidon transversal for sparse blocks). *Let $H = H(n) \rightarrow \infty$, let $\mathcal{B}$ be a family of pairwise disjoint intervals of integers contained in $[n]$, each of length H , and put $M := |\mathcal{B}|$. For each $B \in \mathcal{B}$, let $\mathcal{C}_B \subseteq B$ satisfy*

$$|\mathcal{C}_B| \geq \gamma H$$

for some fixed $\gamma > 0$. Assume that there is a fixed $\alpha > 0$ such that

$$\frac{Mn^\alpha}{H^2} = o(1), \quad \frac{n^\alpha}{H^2} = o(1), \quad \text{and} \quad \frac{\sqrt{n}}{H^2} = o(1).$$

Then there exist choices $c_B \in \mathcal{C}_B$, one for each $B \in \mathcal{B}$, such that

$$C := \{c_B : B \in \mathcal{B}\}$$

is a multiplicative Sidon set.

Proof. Choose independently and uniformly

$$X_B \in \mathcal{C}_B \quad (B \in \mathcal{B}).$$

We shall avoid atomic bad events encoding non-trivial product coincidences.

For distinct blocks B_1, B_2, B_3, B_4 and values $a_i \in \mathcal{C}_{B_i}$, include the atomic event

$$X_{B_i} = a_i \quad (1 \leq i \leq 4)$$

whenever

$$a_1 a_2 = a_3 a_4.$$

For distinct blocks B_1, B_2, B_3 and values $a_i \in \mathcal{C}_{B_i}$, include the atomic event

$$X_{B_1} = a_1, \quad X_{B_2} = a_2, \quad X_{B_3} = a_3$$

whenever either

$$a_1^2 = a_2 a_3 \quad \text{or} \quad a_1 a_2 = a_3^2.$$

Avoiding all these atomic events is sufficient. Indeed, consider any equality

$$c_{B_1} c_{B_2} = c_{B_3} c_{B_4}$$

among selected elements. If a block index occurs on both sides, then cancelling the corresponding positive integer gives either the same unordered pair of block indices, or an equality between selected values from two distinct blocks; the latter is impossible since the blocks are disjoint. We may therefore assume that no block index occurs on both sides.

If the four block indices are distinct, the equality is excluded by one of the four-variable atoms. If exactly one side has a repeated block index, the equality is excluded by one of the square-type atoms. Finally, if both sides have repeated block indices, then the equality has the form $c_B^2 = c_{B'}^2$ with $B \neq B'$, which again forces $c_B = c_{B'}$ and is impossible because the blocks are disjoint. Thus every non-trivial product coincidence is excluded.

We estimate, for a fixed block $B \in \mathcal{B}$, the total probability of all atomic bad events involving X_B . Let

$$U := \bigcup_{B \in \mathcal{B}} \mathcal{C}_B, \quad |U| \leq MH.$$

We use the divisor bound in the form

$$\tau(m) \ll_\alpha m^{\alpha/2} \ll_\alpha n^\alpha \quad (m \leq n^2).$$

First consider four-variable atoms. Up to an absolute factor accounting for the possible positions of X_B , fix a value $a \in \mathcal{C}_B$. After choosing another value $b \in U$, the number of ordered pairs (c, d) with $cd = ab$ is at most $\tau(ab)$. Since the blocks are disjoint, each value of U determines its block. Hence the number of four-variable atoms involving X_B is

$$\ll \sum_{a \in \mathcal{C}_B} \sum_{b \in U} \tau(ab) \ll_{\alpha} M H^2 n^{\alpha}.$$

Each such atom has probability at most $(\gamma H)^{-4}$, so the total four-variable contribution is

$$\ll_{\gamma, \alpha} \frac{M n^{\alpha}}{H^2} = o(1).$$

Next consider square-type atoms. The two estimates below account, up to an absolute factor, for all possible positions of the fixed variable X_B in a three-variable atom. For atoms of the shape $a^2 = bc$ with $a \in \mathcal{C}_B$, the number of possible ordered pairs (b, c) is at most

$$\tau(a^2) \ll_{\alpha} n^{\alpha}.$$

Therefore the total contribution of these atoms is

$$\ll_{\gamma, \alpha} \frac{H n^{\alpha}}{H^3} = \frac{n^{\alpha}}{H^2} = o(1).$$

For atoms of the shape $ab = c^2$ with $a \in \mathcal{C}_B$, write uniquely

$$a = r s^2$$

with r squarefree. Then every positive integer solution has

$$b = r t^2, \quad c = r s t$$

for some positive integer t . Since $b \leq n$, there are at most $O(\sqrt{n})$ possible t . Hence the number of such atoms involving X_B is $O(H\sqrt{n})$, and their total contribution is

$$\ll_{\gamma} \frac{H\sqrt{n}}{H^3} = \frac{\sqrt{n}}{H^2} = o(1).$$

Combining the three estimates, the total probability of all atomic bad events involving any fixed variable is $o(1)$. Since every atomic event depends on at most four variables, this total is at most $1/(8 \cdot 4)$ for all sufficiently large n . Lemma 2.1 therefore applies and gives a choice avoiding all atomic bad events. For this choice, the selected set C is multiplicative Sidon. $\square$

3. COUNTING PRIME-POOR BLOCKS

Fix $0 < \varepsilon < 1/10$ and set

$$\kappa := \frac{10}{33} + \varepsilon, \quad H := \lceil n^{\kappa} \rceil.$$

We shall eventually prove $g(n) \leq 2H$ for all sufficiently large n . Choose a number θ with

$$\frac{10}{33} < \theta < \kappa < \frac{1}{2}.$$

For instance one may take $\theta = 10/33 + \varepsilon/2$. Let

$$B_r := \{(r-1)H+1, (r-1)H+2, \dots, rH\}, \quad 1 \leq r \leq t := \lfloor n/H \rfloor.$$

These are the full blocks of length H .

Fix once and for all a large integer $R \geq 100$. A block B_r is called *bad* if either

$$B_r \cap [1, 2\sqrt{n}] \neq \emptyset$$

or B_r contains at most R primes. All other full blocks are called *good*. Thus every good block lies above $2\sqrt{n}$ and contains at least $R+1$ primes.

Lemma 3.1 (Number of bad blocks). *Let M be the number of bad blocks. Then*

$$M \ll \frac{\sqrt{n}}{H} + 1 + \frac{n^{10/11+o(1)}}{H}.$$

In particular, if $H = n^{10/33+\varepsilon+o(1)}$, then

$$\frac{Mn^{o(1)}}{H^2} = o(1).$$

Proof. There are $O(\sqrt{n}/H + 1)$ bad blocks meeting $[1, 2\sqrt{n}]$. We now count the remaining bad blocks, which are disjoint from $[1, 2\sqrt{n}]$.

Let $B = [u+1, u+H]$ be one of these remaining bad blocks. Then $u+1 > 2\sqrt{n}$, and B contains at most R primes. Since $\theta < \kappa$, we have $n^\theta = o(H)$. Hence, for all sufficiently large n and all

$$x \in [u+1, u+H/2],$$

every integer m with

$$x < m \leq x + x^\theta$$

lies in B . Indeed, $x + x^\theta \leq u + H/2 + n^\theta \leq u + H$ for all large n . Therefore this short interval contains at most R primes. By Lemma 1.6, the contribution of prime powers to $\sum_{x < m \leq x+x^\theta} \Lambda(m)$ is $O((\log x)^2)$, while the contribution of primes is $O_R(\log x)$. Since $x \geq u+1 > 2\sqrt{n}$, this gives

$$\sum_{x < m \leq x+x^\theta} \Lambda(m) = o(x^\theta)$$

uniformly for such x . In particular, for all large n ,

$$\left| \sum_{x < m \leq x+x^\theta} \Lambda(m) - x^\theta \right| \geq \frac{1}{2}x^\theta.$$

Thus each high bad block contributes a subinterval of x -values of measure $\gg H$ to the exceptional set in Theorem 1.3. These subintervals are disjoint as B varies.

For all sufficiently large n , all these x -values are at least $2\sqrt{n}$, so the dyadic intervals to which Theorem 1.3 is applied have lower endpoints tending to infinity. Summing the exceptional-set estimate over dyadic intervals with $X \leq n$ gives total exceptional measure

$$\ll n^{10/11+o(1)}$$

in $[1, n]$. Hence the number of high bad blocks is

$$\ll \frac{n^{10/11+o(1)}}{H}.$$

This proves the asserted bound for M .

Finally,

$$\frac{Mn^{o(1)}}{H^2} \ll n^{1/2-3\kappa+o(1)} + n^{-2\kappa+o(1)} + n^{10/11-3\kappa+o(1)} = o(1),$$

because $\kappa > 10/33 > 1/4$ and $10/11 = 3 \cdot 10/33$. □

4. CHOOSING PRIMES IN THE GOOD BLOCKS

Let $\mathcal{G}$ be the set of good blocks and $\mathcal{B}$ the set of bad blocks. From each good block $G \in \mathcal{G}$, choose a set P_G of exactly R primes contained in G . Every such prime is larger than $2\sqrt{n}$. We choose one prime $q_G \in P_G$ independently and uniformly at random.

Lemma 4.1 (A good choice of the good-block primes). *There is a choice of primes*

$$Q := \{q_G : G \in \mathcal{G}\}$$

such that for every bad block $B \in \mathcal{B}$, at least $H/2$ integers in B are not divisible by any prime in Q .

Proof. Fix a bad block B . For a good block G , let $Y_G(B)$ be the indicator of the event that the randomly chosen prime q_G divides at least one integer of B . The variables $Y_G(B)$ are independent as G varies. Since every $q_G > 2\sqrt{n}$ and every integer $a \leq n$ is divisible by at most one prime larger than $\sqrt{n}$, the total number of pairs

$$(a, q), \quad a \in B, \quad q \in \bigcup_{G \in \mathcal{G}} P_G, \quad q \mid a,$$

is at most H . Consequently

$$\mathbb{E} \sum_{G \in \mathcal{G}} Y_G(B) \leq \frac{H}{R}.$$

Let

$$S_B := \sum_{G \in \mathcal{G}} Y_G(B), \quad \mu_B := \mathbb{E} S_B \leq \frac{H}{R}.$$

By the standard Chernoff bound for sums of independent Bernoulli variables,

$$\mathbb{P} \left(S_B > \frac{H}{2} \right) \leq \left(\frac{e\mu_B}{H/2} \right)^{H/2} \leq \left(\frac{2e}{R} \right)^{H/2} \leq \exp(-cH),$$

with an absolute constant $c > 0$, since $R \geq 100$. Since $H = o(\sqrt{n})$ in the applications below, every chosen prime $q_G > 2\sqrt{n}$ is larger than H . Hence such a prime divides at most one integer in any fixed block of length H . Consequently, for a fixed bad block B , the number of integers in B divisible by at least one chosen prime is at most

$$\sum_{G \in \mathcal{G}} Y_G(B).$$

There are at most n/H bad blocks, and $H \rightarrow \infty$ polynomially. Hence the union bound shows that, with positive probability, every bad block contains at most $H/2$ integers divisible by the chosen primes. This gives the desired deterministic choice of Q . $\square$

Fix such a choice of Q . For every bad block B , define

$$\mathcal{C}_B := \{a \in B : q \nmid a \text{ for every } q \in Q\}.$$

By Lemma 4.1,

$$|\mathcal{C}_B| \geq H/2 \quad (B \in \mathcal{B}).$$

5. COMPLETION OF THE PROOF

Theorem 5.1. *For every $\varepsilon > 0$,*

$$g(n) \ll_{\varepsilon} n^{10/33+\varepsilon}.$$

Equivalently,

$$g(n) \leq n^{10/33+o(1)}.$$

Proof. It is enough to prove the result for all sufficiently large n , since the implicit constant may be enlarged to handle finitely many exceptional values. We may also suppose $0 < \varepsilon < 1/10$. Put

$$\kappa = \frac{10}{33} + \varepsilon, \quad H = \lceil n^{\kappa} \rceil.$$

Use the block decomposition above.

By Lemma 3.1, if M denotes the number of bad blocks, then

$$M \ll \frac{\sqrt{n}}{H} + 1 + \frac{n^{10/11+o(1)}}{H}.$$

Take $\alpha := \varepsilon$. Since $H = n^{\kappa+o(1)}$, we have

$$\begin{aligned} \frac{Mn^\alpha}{H^2} &\ll n^{1/2+\alpha-3\kappa+o(1)} + n^{\alpha-2\kappa+o(1)} + n^{10/11+\alpha-3\kappa+o(1)} \\ &= o(1), \end{aligned}$$

because $\kappa = 10/33 + \varepsilon$, $\alpha = \varepsilon$, and $\kappa > 10/33 > 1/4$. Also

$$\frac{n^\alpha}{H^2} = n^{\alpha-2\kappa+o(1)} = o(1),$$

and

$$\frac{\sqrt{n}}{H^2} = n^{1/2-2\kappa+o(1)} = o(1),$$

because $\kappa > 10/33 > 1/4$. Therefore Lemma 2.2, applied with $\gamma = 1/2$, gives a choice

$$c_B \in \mathcal{C}_B \quad (B \in \mathcal{B})$$

such that

$$C := \{c_B : B \in \mathcal{B}\}$$

is multiplicative Sidon.

Now define

$$A := Q \cup C.$$

Every good block is met by Q , and every bad block is met by C . Hence A meets every full block B_r , $1 \leq r \leq t$.

We next check that A is multiplicative Sidon. Suppose

$$a_1 a_2 = a_3 a_4, \quad a_i \in A.$$

For each $q \in Q$, compare the q -adic valuations of the two sides. The primes in Q are distinct, and no element of C is divisible by any prime in Q by construction. Hence the multiplicity with which q occurs among $\{a_1, a_2\}$ is equal to the multiplicity with which it occurs among $\{a_3, a_4\}$. Therefore the multiset of Q -elements appearing on the left is the same as the multiset of Q -elements appearing on the right. Cancelling this common multiset, we are left with an equality involving only elements of C , with the same number of remaining factors on the two sides. Thus the remaining equality is either $1 = 1$, an equality of single C -elements, or a product equality between two C -elements on each side. In the first two cases it is plainly trivial, and in the last case it is trivial because C is multiplicative Sidon. Thus the original equality is also trivial, and A is multiplicative Sidon.

It remains to convert block coverage into interval coverage. Let $[x, x + 2H] \subseteq [1, n]$. Put

$$r := \left\lfloor \frac{x-1}{H} \right\rfloor + 2.$$

Then

$$x < (r-1)H + 1 \leq rH < x + 2H.$$

Since $x + 2H \leq n$, this gives $rH \leq n$, and hence

$$r \leq \lfloor n/H \rfloor = t.$$

Therefore the full block B_r is contained in $[x, x + 2H]$, and hence this interval meets A .

Thus every interval of length $2H$ contained in $[1, n]$ meets the multiplicative Sidon set A . Consequently

$$g(n) \leq 2H \ll n^\kappa = n^{10/33+\varepsilon}.$$

This proves the theorem. $\square$

Remark 5.2. The exponent $10/33$ enters only through the exceptional-set estimate $\mu(\theta) \leq 10/11$ for $\theta > 10/33$. The local-lemma filling step requires roughly $M \ll H^2$, and the exceptional-set estimate gives $M \ll n^{10/11+o(1)}/H$. Thus one needs

$$\frac{n^{10/11+o(1)}}{H^3} = o(1),$$

which is exactly $H \gg n^{10/33+o(1)}$.

6. CONDITIONAL IMPROVEMENT UNDER THE RIEMANN HYPOTHESIS

The same argument gives a sharper conditional estimate.

Theorem 6.1. *Assume the Riemann hypothesis. Then, for every $\varepsilon > 0$,*

$$g(n) \ll_{\varepsilon} n^{1/4+\varepsilon}.$$

Proof. It is enough to prove the result for all sufficiently large n , and we may suppose $0 < \varepsilon < 1/10$. Put

$$\kappa := \frac{1}{4} + \varepsilon, \quad H := \lceil n^{\kappa} \rceil,$$

and choose

$$\theta := \frac{1}{4} + \frac{\varepsilon}{2}.$$

Then

$$0 < \theta < \kappa < \frac{1}{2}.$$

As before, decompose $[1, n]$ into full blocks

$$B_r := \{(r-1)H + 1, (r-1)H + 2, \dots, rH\}, \quad 1 \leq r \leq t := \lfloor n/H \rfloor.$$

Fix a sufficiently large integer R , and call a block bad if it either meets $[1, 2\sqrt{n}]$ or contains at most R primes. All other full blocks are called good.

We first count the bad blocks. The number of bad blocks meeting $[1, 2\sqrt{n}]$ is

$$O\left(\frac{\sqrt{n}}{H} + 1\right).$$

Now let $B = [u+1, u+H]$ be a bad block disjoint from $[1, 2\sqrt{n}]$. Then $u+1 > 2\sqrt{n}$, and B contains at most R primes. Since $\theta < \kappa$, we have $n^{\theta} = o(H)$. Hence, for all sufficiently large n and all

$$x \in [u+1, u+H/2],$$

every integer m with

$$x < m \leq x + x^{\theta}$$

lies in B . Therefore this short interval contains at most R primes. By Lemma 1.6, the contribution of prime powers to

$$\sum_{x < m \leq x+x^{\theta}} \Lambda(m)$$

is $O((\log x)^2)$, while the contribution of primes is $O_R(\log x)$. Since $x \geq u+1 > 2\sqrt{n}$, we get

$$\sum_{x < m \leq x+x^{\theta}} \Lambda(m) = o(x^{\theta})$$

uniformly for such x . Therefore each high bad block contributes a set of x -values of measure $\gg H$ to the exceptional set in Theorem 1.5. These sets are disjoint as the block varies.

For all sufficiently large n , the relevant x -values are at least $2\sqrt{n}$, so the lower endpoints in the dyadic decomposition tend to infinity. Summing Theorem 1.5 over dyadic intervals gives total exceptional measure

$$\ll n^{1-\theta+o(1)}$$

in $[1, n]$. Hence, if M denotes the number of bad blocks, then

$$M \ll \frac{\sqrt{n}}{H} + 1 + \frac{n^{1-\theta+o(1)}}{H}.$$

Taking $\alpha := \varepsilon$, the above bound for M gives

$$\begin{aligned} \frac{Mn^\alpha}{H^2} &\ll n^{1/2+\alpha-3\kappa+o(1)} + n^{1-\theta+\alpha-3\kappa+o(1)} \\ &= o(1), \end{aligned}$$

because

$$3\kappa = \frac{3}{4} + 3\varepsilon, \quad 1 - \theta = \frac{3}{4} - \frac{\varepsilon}{2}, \quad \alpha = \varepsilon.$$

Also

$$\frac{n^\alpha}{H^2} = n^{\alpha-2\kappa+o(1)} = o(1),$$

and

$$\frac{\sqrt{n}}{H^2} = n^{1/2-2\kappa+o(1)} = n^{-2\varepsilon+o(1)} = o(1).$$

Choose primes in the good blocks exactly as in Lemma 4.1. The same proof applies here, since $H = o(\sqrt{n})$ and every selected good-block prime is larger than $2\sqrt{n}$. Thus we obtain a set Q of one prime from each good block such that, for every bad block B , the set

$$\mathcal{C}_B := \{a \in B : q \nmid a \text{ for every } q \in Q\}$$

satisfies

$$|\mathcal{C}_B| \geq H/2.$$

By Lemma 2.2, we may choose one element $c_B \in \mathcal{C}_B$ from each bad block so that

$$C := \{c_B : B \text{ bad}\}$$

is multiplicative Sidon.

Set

$$A := Q \cup C.$$

Then A meets every full block. The same valuation argument used in the proof of Theorem 5.1 shows that A is multiplicative Sidon. Finally, the same block-covering argument gives that every interval of length $2H$ contained in $[1, n]$ contains one full block and hence meets A . Therefore

$$g(n) \leq 2H \ll n^{1/4+\varepsilon},$$

as claimed. $\square$

Remark 6.2. The exponent $1/4$ is the natural endpoint of the present filling step. Indeed, even if the exceptional-set estimate were stronger, Lemma 2.2 still requires

$$\frac{\sqrt{n}}{H^2} = o(1),$$

which comes from square-type product coincidences of the form $ab = c^2$. Thus this version of the argument cannot by itself go below $n^{1/4+o(1)}$.

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